

HiMe: Hierarchical Embodied Memory for Long-Horizon Vision-Language-Action Control

Anonymous Authors¹

Abstract

Current Vision-Language-Action (VLA) models excel at robotic manipulation but often struggle with non-Markovian tasks requiring long-term memory and reasoning due to their reliance on immediate observations. Existing solutions face a “frequency-competence paradox,” where high-performance models are too slow for real-time control, while faster models lack sufficient reasoning capabilities. To resolve this architectural misalignment, we propose **HiMe**, a Hierarchical Embodied Memory framework that decouples embodied intelligence into a high-frequency Executor for execution, a Sentry for working memory, and a Planner for long-term strategy. We also introduce a dynamic knowledge system based on cross-modal semantic schemas and active management mechanisms, allowing robots to maintain memory plasticity through “Add, Update, and Delete” operations. This hierarchical design effectively balances the conflict between real-time execution and slow thinking planning, significantly improving success rates in long-horizon tasks. Experiments demonstrate that this approach not only outperforms flat memory baselines but also exhibits the novel ability to self-correct its internal knowledge based on human preferences.

1. Introduction

Vision-Language-Action (VLA) models have emerged as a powerful paradigm for general-purpose robotic manipulation with large-scale internet-level pretraining. By directly mapping observations to control signals, they provide a powerful foundation for executing complex motor skills. Most existing architectures rely on the Markov assumption, where the policy $p(a_t|o_t, l)$ is conditioned only on the transient

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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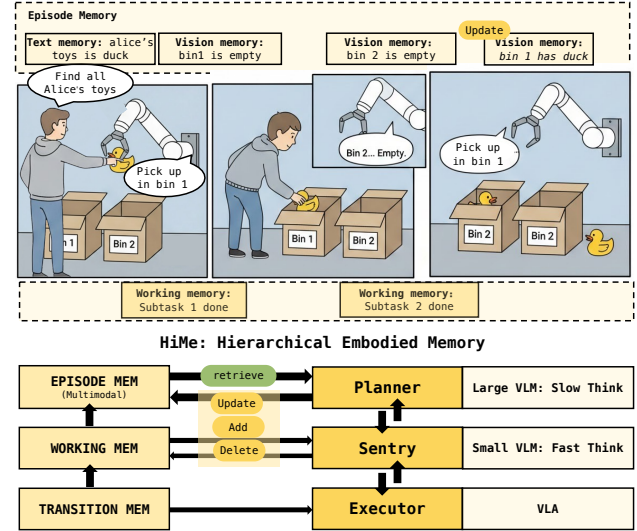


Figure 1. **Overview of HiMe.** A motivating example (top) illustrates the need for maintaining and updating task-relevant multi-modal memory across long-horizon subtasks. HiMe addresses this challenge by organizing embodied intelligence into a hierarchical structure that separates fast execution from memory-driven reasoning over long time scales.

observation at the current time step. This inherent limitation prevents them from maintaining a persistent belief of the environment in non-Markovian settings. Consequently, VLAs struggle with complex, long-horizon tasks that demand long-range temporal dependencies, where the optimal action depends on a chain of past events or latent information that is no longer visible in the current sensory stream.

To overcome these limitations, one intuitive approach is to imbue VLA models with native memory capabilities through specialized training or auxiliary losses (Torre et al., 2025; Fang et al., 2025). However, these training-based methods are often constrained by limited context windows and the inherent difficulty of optimizing long-range causal dependencies over hundreds of steps. Another alternative is to introduce Large Vision-Language Models (VLMs) as high-level memory containers to store and retrieve historical trajectories (Sridhar et al., 2025). Yet, real-time robotic control imposes a strict upper bound on the deployable VLM scale, as control demands high-frequency, low-latency execution.

This scale constraint, in turn, limits the internal world knowledge and generalization capabilities of the VLM, weakening its zero-shot performance. Importantly, most memory operations do not intrinsically require high-level reasoning or broad generalization.

Motivated by this temporal and scale mismatch, we decouple memory and reasoning operations into hierarchical layers with distinct time scales. We propose **HiMe**, a **Hierarchical Embodied Memory** framework that decouples embodied intelligence into three functional layers with distinct temporal resolutions, mirroring the multi-store structure of human cognition. (1) The **Executor (VLA)** governs **transient memory**, focusing on high-frequency sensory-motor coordination for physical stability. (2) The **Sentry** acts as the guardian of **working memory** (short-term memory); it asynchronously filters the continuous sensory stream to identify critical state transitions, effectively bridging the gap between raw perception and semantic events. (3) The **Planner** manages **episodic memory** (long-term memory). By invoking the “slow-thinking” Planner only at the discrete junctions identified by the Sentry, our architecture preserves deep strategic reasoning while maintaining real-time execution, effectively resolving the frequency-capability paradox.

However, a static memory hierarchy alone is insufficient for the complexities of real-world human-robot interaction, which is characterized by: (1) *multimodal richness*, where human instructions carry dense logical constraints and latent preferences—such as designating a specific red cup as “Alice’s cup”—that are nearly impossible to capture through vision-only trajectories; and (2) *high dynamism*, where task goals and environment states are not static but evolve. A memory system that only supports passive accumulation (Sridhar et al., 2025) inevitably suffers from knowledge stagnation and cognitive dissonance when faced with such outdated or conflicting experiences.

Furthermore, our Hierarchical Memory framework treats memory as a multimodal, dynamic knowledge system centered around two fundamental dimensions:

(1) *What to memorize?: Cross-modal Semantic Schemata.* To bridge the gap between raw perception and logical intent, we represent episodic memory as an object-centric system where visual experiences are deeply anchored with high-density textual descriptions. This organization allows the Planner to retrieve not only visual features but also the underlying ownerships, procedural rules, and human preferences that are invisible in pure pixel space.

(2) *How to memorize?: Active Management Mechanism.* In contrast to passive storage, we introduce explicit *Add*, *Update*, and *Delete* operations to grant the robot knowledge plasticity. When the Sentry detects a significant state transition or a revision in user intent, the Planner proac-

tively refines the knowledge base. By updating outdated beliefs and purging redundant or erroneous data, the agent maintains a consistent and concise memory that evolves in alignment with the environment.

We evaluate our method across a suite of challenging long-horizon manipulation tasks that require multi-step reasoning and adaptation to shifting human preferences. Experimental results show that our approach significantly outperforms existing flat-memory baselines in both success rate and computational efficiency. Notably, our agent demonstrates the ability to “self-correct” its internal knowledge base when faced with conflicting instructions, a capability that is absent in prior retrieval-based methods.

Our core contributions are summarized as follows:

- We propose a **Hierarchical Memory Management framework** that decouples robotic control into transient (Executor), working (Sentry), and episodic (Planner) memory layers, resolving the granularity conflict in long-horizon VLA tasks.
- We introduce a Cross-modal Memory organization (*what to memorize*) combined with an Active Management mechanism (*how to memorize*), enabling the robot to maintain knowledge plasticity and alignment in highly dynamic and multimodal interactions.
- We provide **extensive empirical evidence** demonstrating that our hierarchical, self-evolving memory architecture achieves superior performance and robustness in complex human-robot collaborative scenarios with significantly reduced VLM computational overhead.

2. Related Work

2.1. Foundation models in robotics

Recent advancements in End-to-End Vision-Language-Action (VLA) models adopt a co-training paradigm that integrates a pre-trained VLM backbone with a dedicated action expert (Black et al., 2024; Intelligence et al., 2025). By harnessing the VLM’s extensive world knowledge (Bai et al., 2025) to interpret complex scenes, this approach effectively transforms the backbone into a generalist agent capable of mapping raw observations directly to continuous actions. Alternatively, the hierarchical paradigm decouples reasoning from execution (Shi et al., 2025b; Li et al., 2025b; Shentu et al., 2024). In this framework, a high-level policy, typically a VLM, processes visual observations and task descriptions to generate intermediate goals, such as natural language sub-tasks or latent embeddings. These intermediate representations then condition a low-level policy to output continuous actions. Building upon this hierarchical structure, our work introduces a novel hierarchical memory

architecture designed to empower the high-level VLM to generate more precise and context-aware instructions for the low-level controller.

2.2. Robotic memory

Memory capabilities are fundamental for VLA to successfully execute complex, long-horizon manipulation tasks. Consequently, extensive research has been dedicated to endowing vla with historical context, though most existing implementations remain predominantly image-based (Huang et al., 2022). These works typically follow two paradigms: utilizing external memory banks (Li et al., 2025a; Torne et al., 2025; Fang et al., 2025; Huang et al., 2023; Zhu et al., 2020; Zheng et al., 2024; Hu et al., 2025), that employ FIFO queues (Sridhar et al., 2025) or fused buffers (Shi et al., 2025a) to store raw observations, or exploring history compression to distill information into compact representations. Notably, MemER (Sridhar et al., 2025) recently advanced the latter by proposing keyframe selection algorithms to better preserve critical information within subtasks. Distinct from these methods, we propose a cross-modal, explicitly managed memory bank designed to enhance task consistency and effectively retain long-term temporal context.

3. Methods

3.1. Overview

Long-horizon manipulation relies on two contradictory capabilities: maintaining global consistency over extended history and reacting responsively to immediate dynamics. End-to-end approaches typically struggle to balance these needs—short-context policies suffer from catastrophic forgetting, while heavy reasoning models incur prohibitive latency for real-time control.

To address this challenge, we propose HiMe, a Hierarchical Embodied Memory framework that decouples embodied intelligence into three functional layers: (1) The Executor π_e : A high-frequency VLA that ensures physical stability by mapping immediate observations to actions. (2) The Sentry π_s : A lightweight VLM that identifies progress changes in the environment, triggering planning only when necessary. (3) The Planner π_p : A heavyweight VLM that uses long-term episodic memory and plans at a lower frequency. This separation allows the system to amortize the cost of complex planning over extended execution windows, ensuring the low-level controller remains responsive to immediate physical dynamics while maintaining long-term task coherence.

Problem Setup. We formulate the long-horizon manipulation task as a sequential decision-making process under partial observability. The agent operates over a horizon $T \gg 1$, receiving high-dimensional observations $o_t \in \mathcal{O}$ and a natural language instruction l . The objective is to

generate an action sequence $a_{0:T}$ that transitions the environment to a state satisfying the instruction l . Ideally, the optimal action at any step t depends on the full interaction history $h_t = (o_0, a_0, \dots, o_t)$ to resolve ambiguity and track progress. However, processing h_t continuously is computationally intractable. To address this, we adopt a hierarchical policy $\pi(a_t \mid o_{0:t}, l)$.

3.2. Hierarchical Embodied Memory

To bridge the gap between reflexes and reflection, we augment the agent with a structured state space comprising three distinct components. This explicitly represents the information required for immediate control, temporal monitoring, and long-term planning.

Transient Memory ($\mathcal{T}_t$). Managed by the Executor, it contains the instantaneous observation o_t . It serves as the minimal sufficient state for reactive sensory-motor coordination.

Working Memory ($\mathcal{W}_t$). To support high-frequency monitoring, we maintain a sliding window of recent observations $\mathcal{W}_t = \{o_{t-h_s}, \dots, o_t\}$. Unlike the main memory, this buffer is transient and raw, capturing the immediate dynamics required for verifying subgoal status or detecting sudden failures.

Episodic Memory ($\mathcal{E}_t$). A persistent long-term store managed by the Planner, consisting of: (1) *Contextual Memory* $\mathcal{E}_t^c$, a multimodal key-value store $\mathcal{M}_t$ that accumulates compact, task-relevant information—including both visual imagery and textual descriptions—over time, such as object state changes, spatial constraints, or human preferences, and supports retrieval via textual keys. (2) *Procedural Memory* $\mathcal{E}_t^p$, a structured plan $\mathcal{E}_t^p = \{\tau_i\}_{i=1}^N$ consisting of an ordered list of language subgoals, where the active subgoal $\tau_t^{(i)}$ carries status labels like ‘active’, ‘pending’ or ‘done’ to synchronize the control loop.

3.3. Hierarchical Policy Decomposition

We decompose the hierarchical policy $\pi(a_t \mid o_{0:t}, l)$ into a three-tiered hierarchy. This design ranges from a memory-free execution layer to a long-horizon reasoning layer, progressively trading off frequency for context capacity, decoupling immediate actuation from high-level reasoning. Formally, the decision process at time t involves interactions between a base executor π_e , a monitoring Sentry π_s , and a reasoning Planner π_p .

Base-level Executor (π_e). At the base level, the executor functions as a high-frequency reactive policy. We assume a local Markov property where, given an active language subgoal $\tau_t \in \mathcal{E}_t^p$, the optimal action is conditionally independent of the long-term history. The executor maps the

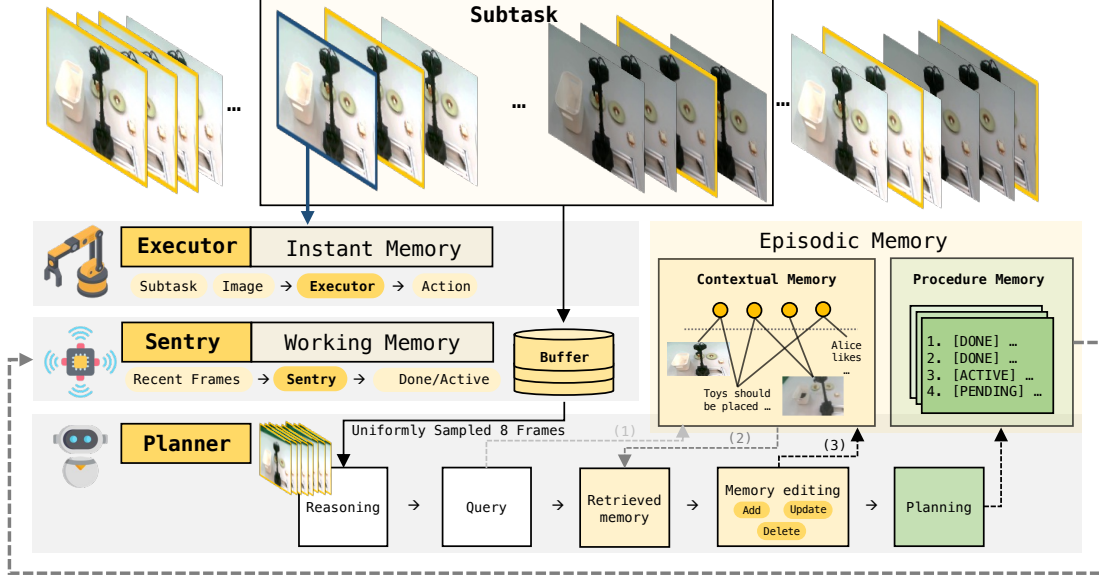


Figure 2. **Architecture of HiMe.** The Sentry module serves as a real-time monitor, periodically tracking the executor’s progress and buffering visual observations. Upon detecting the completion of a subtask, it triggers the Planner to review the entire execution trajectory. The Planner then performs memory retrieval and consolidation to update the contextual memory and refines the procedure memory. Finally, the refined instruction is fed back to the Sentry to resume monitoring the executor, forming a robust closed-loop system.

current observation and subgoal to an action:

$$a_t \sim \pi_e(a_t \mid o_t, \tau_t), \quad (1)$$

where τ_t serves as a conditioning variable abstracted from the complex episodic history, representing the current executing subgoal. This design ensures that π_e remains stateless and computationally efficient, enabling real-time responsiveness to physical dynamics.

Agile Sentry (π_s). The second layer is the Sentry, it is a gating function $u_t \in \{0, 1\}$ that be used to amortize the computational cost of reasoning. Operating on the sliding window working memory $\mathcal{W}_t$, the Sentry evaluates subgoal completion or environmental deviation at intervals of n_m :

$$u_t = \begin{cases} \mathbb{I}[\pi_s(\mathcal{W}_t, \tau_t) > \delta] & \text{if } t \equiv 0 \pmod{n_m} \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where $\mathbb{I}[\cdot]$ is the indicator function and δ is a decision threshold. $u_t = 1$ signals a “handover” event—indicating that the current subgoal τ_t is either completed or invalidated, thereby triggering the Planner. When $u_t = 0$, the system bypasses high-level reasoning, maintaining the current latent state $\mathcal{E}_{t+1} \leftarrow \mathcal{E}_t$.

Reasoning Planner (π_p). When triggered ($u_t = 1$), the Planner performs a belief update on the episodic memory $\mathcal{E}_t$ to re-align the agent’s internal state with the environment. This process is formulated as a two-stage transition:

(1) *Context Retrieval:* The Planner first generates a query q to retrieve a subset of task-relevant knowledge $\mathcal{M}_{ret} \subset \mathcal{E}_t^c$

via semantic similarity:

$$q = f_{enc}(l, \mathcal{W}_t); \quad \mathcal{M}_{ret} = \text{TopK}(\mathcal{E}_t^c, q). \quad (3)$$

(2) *Memory Update & Re-planning:* Conditioned on the global instruction l , working memory $\mathcal{W}_t$, and retrieved context $\mathcal{M}_{ret}$, the Planner updates the episodic memory state:

$$(\mathcal{E}_{t+1}^c, \mathcal{E}_{t+1}^p) \leftarrow \pi_p(l, \mathcal{W}_t, \mathcal{M}_{ret}, \mathcal{E}_t^p). \quad (4)$$

Specifically, this update involves symbolic operations on the Contextual Memory $\mathcal{E}^c$ to reflect new world states, and the synthesis of a new procedural plan $\mathcal{E}_{t+1}^p$. This involves three atomic operations: (a) **Add**: registering new environmental facts or user preferences; (b) **Update**: rectifying outdated entries; and (c) **Delete**: removing stale or conflicting data to maintain a concise representation. The next subgoal τ_{t+1} is then sampled from the updated plan, closing the hierarchical loop.

3.4. Implementation Details

While our framework is model-agnostic, we instantiate the Planner with GPT-4o (Hurst et al., 2024) for its strong multimodal reasoning capabilities, and the Sentry with Qwen3-VL-8B (Bai et al., 2025) for its balance of speed and grounding accuracy. The memory backend utilizes a vector database for semantic retrieval, utilizing OpenAI’s text-embedding-3 model to encode text queries and storing multimodal memory entries alongside their caption embeddings for cosine similarity retrieval. We set the buffer

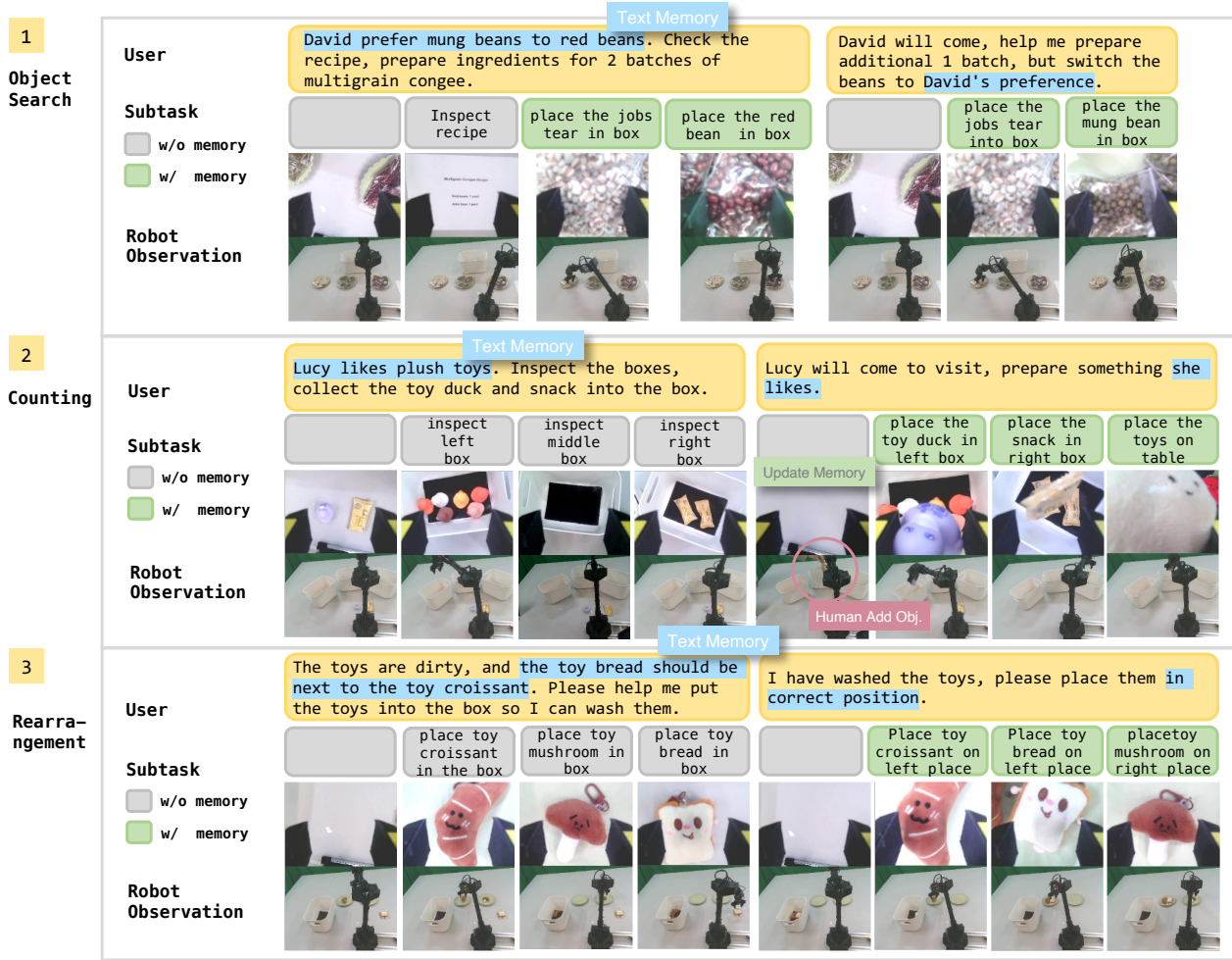


Figure 3. Tasks used in our evaluations. We report performance across 20 trials per task per method.

size $h_s = 8$ and the monitoring interval $n_m = 5$. Consistent with our design philosophy, only the lightweight π_e , a fine-tuned VLA based on $\pi_{0.5}$, requires domain-specific training. The Sentry and Planner operate in a zero-shot or few-shot manner, leveraging pre-trained generalization to handle long-horizon logic without extensive data collection.

4. Experiments

4.1. Task Setting

To demonstrate the critical role of hierarchical memory in long-horizon tasks, we designed three distinct tabletop manipulation scenarios, as illustrated in Fig. 3. These tasks are specifically curated to evaluate diverse capabilities, including user interaction, preference memory, updating memory and planning with exploration.

Object Search. We design a Domestic Maintenance task to evaluate the robot’s capability to integrate inspection, sorting, and preference recall. In this scenario, the robot

must first inspect opaque boxes to deduce storage rules and organize scattered items accordingly. Crucially, during this process, the user introduces new toys into the boxes to test the robot’s ability to dynamically update its memory. Finally, the robot faces a retrieval challenge, where it must synthesize memorized user preferences with the box contents identified during inspection to retrieve the correct object.

Counting. In this scenario, the robot is required to interpret a visual recipe on the table and plan its subsequent actions. The task proceeds in stages: the robot must first *inspect* the recipe to extract the ingredient composition and prepare the materials based on the number of servings requested by the user. Following this, the user introduces a specific preference. To fulfill this customized demand, the robot must synthesize the new constraint with the previously memorized recipe to dispense the correct personalized ingredients.

Rearrangement. In this playroom scenario, the robot performs a two-stage “clear and restore” operation. Initially,

the robot collects all toys scattered on the table into a storage box for cleaning. After a temporal interval, the robot is tasked with restoring the items to the environment. Crucially, the user’s specific placement preference is provided beforehand as prior knowledge. Therefore, to execute the restoration, the robot must recall this pre-established preference and synthesize it with its visual memory of the original spatial configuration to plan the final placement of each item.

Evaluation Setup We use a WidowX-250s arm with a parallel gripper and dual-camera visual input (third-person and wrist view). The Executor π_e runs at 2 Hz, predicting an action chunk A_t of 10 actions (10 Hz), of which 5 are executed open-loop. To monitor the subtask progress, the Sentry π_s is queried after every 10 execution steps of π_e . The Sentry will trigger Planner π_p if the current subtask is completed for slow thinking. Each task’s evaluation metrics are detailed in App. A.

4.2. Main Experiment

To thoroughly validate the effectiveness of our *hierarchical memory* and the *monitor-based* trigger mechanism, we keep the low-level policy π_l and the planner backbone constant, varying only the **memory context** and the **planning frequency**. The comparisons are designed as follows:

Transient Memory: A standard hierarchical VLA baseline where a memory-less planner is invoked at fixed intervals (n_m) and observes only the current frame o_t . This represents the current paradigm of hierarchical robot foundation models, such as Hi-robot (Shi et al., 2025b).

Transient Memory w/ Sentry: This variant introduces our Sentry module to trigger the planner based on task progress. However, the planner remains memory-less, receiving only o_t upon being triggered. It evaluates the benefit of the more stabilized planning from the sentry’s gating.

Flat Memory: The planner operates at a fixed frequency and receives the 8 most recent observations, assisted by a FIFO queue of historical keyframes it previously selected. This setting tests whether unstructured, flat memory is sufficient for long-horizon tasks. This setting is similar to MemER (Sridhar et al., 2025).

HiMe w/o Sentry: We utilize our complete planner’s memory design but remove the sentry module. The planner is forced to query the VLM at a fixed frequency regardless of subtask progress. This ablation validates the sentry’s role in reducing computational redundancy and aligning planning steps with the dynamics of environments.

HiMe (Ours): The proposed framework, where the Sentry dynamically triggers planning based on subtask progress, and the planner utilizes the full hierarchical memory to

Table 1. Comparison of hierarchical settings in the main experiment.

Hierarchy Setting	Trigger	Planner Memory
Transient Memory	Periodic	Current observation
Transient Memory w/ Sentry	Sentry	Current observation
Flat Memory	Periodic	Recents + FIFO
HiMe w/o Sentry	Periodic	Recents + keyframes
HiMe (Ours)	Sentry	Recents + keyframes
Human High-level	Sentry	Human oracle

ensure global consistency.

Human High-level: A human oracle provides the correct subtask description at each step. This serves as an upper bound on task performance given the fixed capabilities of π_l .

4.3. Analysis of Sentry Mechanism

We analyze the Sentry mechanism’s contribution from two key dimensions: *execution consistency* and *observation quality*.

1) *Consistency via Reduced Task Switching:* Even in a Planner with transient memory setting, the Sentry significantly boosts performance. In Fig. 4, we observe a substantial improvement (14% vs. 26%) when comparing *Transient Memory* with *Transient Memory w/ Sentry*. Without the Sentry, the planner operates on a frame-by-frame style, and this high-frequency re-evaluation makes the robot hypersensitive to transient visual noise, leading to frequent, erratic switching between subtasks. The Sentry prevent the planner from intervening until the current subtask is completed, reducing unnecessary task switching and ensures that actions are executed coherently. It reveals that **the core value of the Sentry here is enforcing temporal consistency of Planner.**

2) *Quality via Uniform Sampling vs. Recent Frames:*

The Sentry’s ability to consolidate observations into high-quality working memory largely explains the gap between *HiMe w/o Sentry* and *HiMe*, with average task progress rising from 68% to 90%. Whereas standard planners rely on a narrow window of recent frames, Sentry aggregates the history of the active subtask into structured working memory. This provides a global perspective for success verification, and the consolidated memory allows the Planner to verify the entire progression, effectively reducing hallucinations and redundant replanning.

4.4. Analysis of Planner Memory Management

We validate the necessity of our hierarchical design by comparing *HiMe* against memory-free baselines and flat contextual memory approaches.

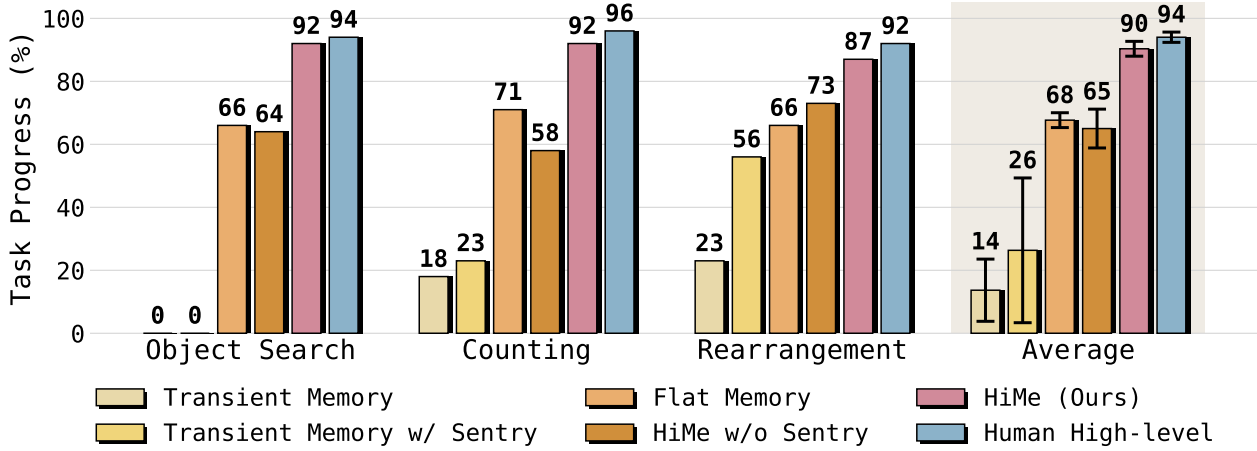


Figure 4. Main Results. We compare HiMe against Transient, Sentry, and Flat Memory baselines across three long-horizon tasks. HiMe significantly outperforms all baselines, achieving a 90% average success rate. Notably, our method effectively bridges the gap between robot and the Human High-level oracle.

(1) *The Necessity of Memory:* Long-horizon tasks require persistent state tracking. In *Counting*, even with Sentry stabilization (*Transient Memory w/ sentry*), the robot still fails (23%) due to the lack of persistence: without an explicit memory management, it cannot retain states (e.g., counted objects) once they leave the transient observable state.

(2) *Consolidation vs. FIFO Queues:* A critical insight from Fig. 4 is the inefficiency of *Flat Memory*, which functions as a contextual memory based on FIFO queues. Although it stores history, it achieves a significantly lower task progress (65%) compared to *HiMe* (90%). The fundamental limitation of such contextual memory is the lack of consolidation: a simple FIFO queue cannot distinguish between truly critical frames and redundant observations. In long-horizon tasks, the limited context window is quickly flooded with noisy frames, causing earlier critical information to be discarded. *HiMe* addresses this by actively consolidating memory at the subtask level. This ensures that essential high-level information is preserved regardless of the episode length, enabling high-precision retrieval with minimal planner overhead.

5. Further Analysis

5.1. Q1: What Type of Memory Representation is Most Effective for Robotic Tasks?

We investigate the impact of memory modality by comparing purely text memory (*Only text*), purely visual memory (*Only image*), and our cross-modal approach.

1) *Spatial and Recognition Demands in Robotics:* Our study results show that robotics tasks impose strong spatial localization and fine-grained recognition demands that caption only memory is inadequate for dynamical physical interac-

tions. Text is a lossy compression: if perception initially misses an object or its spatial context, discarding raw visuals prevents visual re-grounding to recover details or correct errors. In *Object Search*, *Only image* (86%) substantially outperforms *Only text* (74%), which shows that memory with visual evidences is significantly stronger.

2) *Text for Semantics and Logic:* Conversely, text memory is indispensable for non-spatial reasoning. It efficiently summarizes semantics, captures logical dependencies like task sequencing, and retains user preferences. In tasks requiring semantic reasoning, such as *Counting*, *Only text* (91%) outperforms *Only image* (78%), since these semantic clues are hard to extract from pixels on demand.

3) *Superiority of Interleaved Memory:* Our cross-modal approach achieves the best performance (Average 90%) by combining the spatial/recognition fidelity of images with the semantic and logical structure of text, enabling our hierarchical memory system can perform both precise grounding and effective reasoning.

5.2. Q2: Are Dynamic Memory Management Necessary?

To assess the need for dynamic memory management, we run an ablation against two baselines: (1) *No Management*, which only supports Add/Retrieve and keeps all observations as an append-only buffer; and (2) *FIFO*, which caps memory at 8 entries and evicts the oldest when full.

1) *The Cost of Forgetting:* In Fig. 6, *FIFO* performs worst (68% average), far below *No Management* (86%). This shows that for long-horizon tasks, **early context is critical**: naive eviction can remove essential information (e.g., an object location observed early) needed for later reasoning.

Table 2. Comparison of different methods. We report API calls, memory hit, and average scores across three tasks. Here, API Calls refers to the average Planner requests per subtask, and Memory Hit measures the presence of required information in memory during memory-dependent subtask.

Method	Components			Search & Retrieval			Counting			Clear & Restore		
	Sentry	Memory Management	Memory Size	API Call (↓)	Memory Hit (↑)	Avg. Score (↑)	API Call (↓)	Memory Hit (↑)	Avg. Score (↑)	API Call (↓)	Memory Hit (↑)	Avg. Score (↑)
HiMe (Ours)	✓	✓	<i>infinite</i>	1.8	94%	92%	2.6	98%	92%	1.4	92%	87%
Keyframe Memory	×	×	<i>recent 8</i>	5.4	68%	64%	4.8	61%	58%	6.2	76%	73%

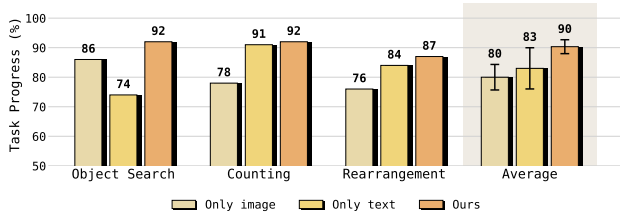


Figure 5. **Ablation of Modality.** HiMe consistently outperforms both text-only and image-only baselines across all three tasks, verifying the robustness of our cross-modal memory mechanism.

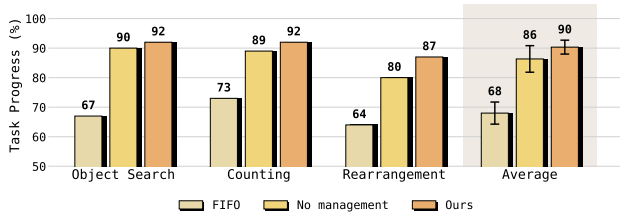


Figure 6. **Ablation of Management.** Comparison between FIFO, No management, and our approach. HiMe consistently achieves the highest task progress across all three tasks, proving the effectiveness of our memory retrieval and consolidation mechanism in long-horizon tasks.

2) *The Value of Consistency:* While *No Management* achieves decent performance by retaining all history, it is still inferior to our full method (86% vs. 90%). Although *No Management* retains all history, it remains below our full method (86% vs. 90%) due to **redundancy and inconsistency**. Without *Update/Delete*, memory stores obsolete states (e.g., prior object locations) alongside current ones, introducing noise during *Query* and potentially confusing the planner.

3) *Necessity of Active Management:* Our method performs best by actively curating memory: *Update* refreshes outdated entries to maintain consistency, and *Delete* removes redundancy. Thus, storing more data is insufficient; effective robots must maintain a concise, consistent memory.

5.3. Q3: Why Does HiMe Achieve Lower Latency and Fewer API Calls?

We further analyze efficiency and reliability by recording total Planner invocations (average *API Calls* per subtask)

and the *Memory Hit* rate (whether required historical context is available at query time).

Beyond the gains in Sec. 4.3 and Sec. 4.4, Table Tab. 2 shows a clear efficiency advantage: *HiMe* reduces API calls by $\sim 3\times$ (e.g., 5.4 to 1.8 in Search & Retrieval) over Flat memory. Since VLM inference dominates latency, fewer calls directly yield faster execution. This efficiency stems from two structural advantages over the keyframe memory:

1) *Sentry Reduces Invocation Frequency:* In keyframe methods, the Planner is often queried every step to interpret the visual buffer, placing the expensive LLM in a high-frequency control loop. Our Sentry instead handles *subtask completion checking* after the Planner specifies a subtask, it runs the dense verification loop and wakes the Planner only when necessary. Thus, Planner involvement shifts from *step-by-step* to sparse *subtask-level* calls.

2) *Infinite Memory vs. FIFO Forgetting:* Flat Memory uses a FIFO buffer (Tab. 1), forming a sliding window that forgets early observations. The resulting low memory hit rate (68% vs. 94%) forces redundant exploration when needed context has been evicted. In contrast, HiMe maintains an infinite structured memory, avoiding this forgetting and enabling immediate retrieval without physical re-exploration.

6. Conclusion

In this work, we presented HiMe, a novel hierarchical embodied memory framework that resolves the fundamental frequency-competence paradox in long-horizon Vision-Language-Action control. By decoupling embodied intelligence into a high-frequency Executor, a progress-aware Sentry, and a strategic Planner, we mirror the multi-store structure of human cognition to balance real-time responsiveness with deep reasoning. Our introduction of cross-modal semantic schemata and active management mechanisms (Add, Update, Delete) transforms robotic memory from a passive observation buffer into a dynamic, self-evolving knowledge system. Extensive experiments demonstrate that HiMe achieves a state-of-the-art 90% average success rate—effectively bridging the gap to human-level performance. We believe that this shift from flat, transient policies to hierarchical, structured memory is a vital step toward developing truly autonomous and adaptive robotic agents.

Impact Statement

This paper presents work where the goal is to advance the fields of machine learning and robotics. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Details of Evaluation Metrics

Object Search We evaluate the robot’s proficiency by monitoring whether it performs necessary inspections, correctly memorizes box contents, places items in the appropriate containers, updates the memory with the latest observation and successfully retrieves the target object.

Counting We assess the robot’s performance based on three criteria: (1) whether the robot actively and correctly inspected the visual recipe; (2) the accuracy of the ingredient preparation relative to the recipe and requested servings; and (3) whether the robot succeed in satisfying the final customized user preference.

Rearrangement We measure task success based on two criteria: (1) the completeness of the initial collection into the storage box; and (2) the accuracy of the final restoration, specifically whether items were placed in the correct location.

B. Additional Related Work

B.1. Memory management in Agents

Effective memory management is crucial for handling long-context interactions. Existing frameworks typically organize memory using hierarchical structures (Liu et al., 2025) or OS-inspired interfaces (Packer et al., 2023), often relying on explicit management strategies (Chhikara et al., 2025; Xu et al., 2025; Yan et al., 2025) to maximize efficiency. A dominant strategy among these is periodic summarization (Long et al., 2025; Yeo et al., 2025), where recent history is compressed into long-term storage at fixed temporal intervals to decouple storage from reasoning. However, these methods have predominantly focused on text-based representations for LLMs (Liang et al., 2023). With the rapid evolution of Vision-Language Models (VLMs), there is a growing necessity to incorporate multimodal information. Our approach addresses this by constructing a cross-modal memory that synergizes text and images, while simultaneously introducing a structured management mechanism tailored to effectively utilize this rich context for complex robotic tasks.

C. Model Initialization and Hyperparameters

Training the Low-Level Policy The hyperparameters for fine-tuning the $\pi_{0.5}$ low-level policy are detailed in Table 3. The model is fine-tuned from the public $\pi_{0.5}$ checkpoint trained on the DROID dataset (Khazatsky et al., 2024).

Table 3. Hyperparameters for Low-Level Policy ($\pi_{0.5}$) fine-tuning.

Hyperparameter	Value
Optimizer	AdamW
β_1	0.9
β_2	0.999
Weight Decay	0.1
Gradient Clip Norm	1.0
LR Schedule	Cosine
Warmup Ratio	0.01
Batch Size	256
Training Steps	30000
Compute	160 H100 GPU hours

C.1. Q6: Analysis of Sentry’s Working Memory and Termination Policy

To understand the Sentry’s decision-making boundary, we analyzed its performance on subtask completion detection under different working memory sizes (window length N). Fig. 7 presents the Precision and Recall, where Subtask “Done” is treated as the positive class.

1) Benefits of Temporal Context: As shown in the Fig. 7, increasing the context window from 1 to 8 frames improves both Precision (from $\sim 76\%$ to $\sim 82\%$) and Recall (from $\sim 22\%$ to $\sim 35\%$). A single frame often lacks sufficient information to distinguish between a temporary pause and true completion. By observing a sequence of 8 frames (consistent with our sentry’s working memory), the Sentry can leverage temporal cues to make more reliable judgments.

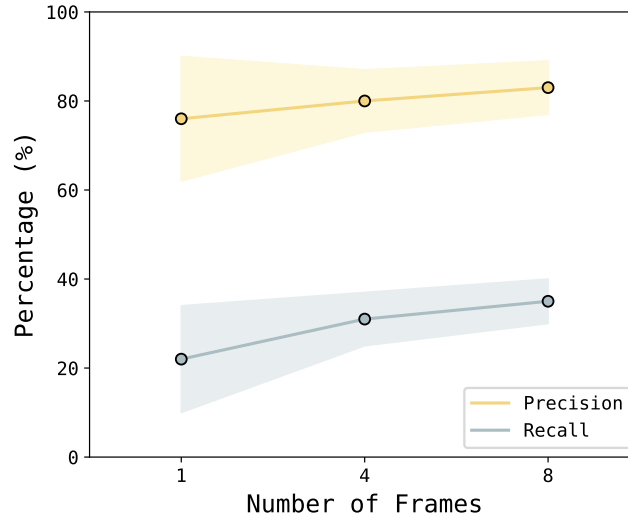


Figure 7. Increasing the number of recent frames provided to the Sentry leads to a consistent improvement in both precision and recall. This indicates that a longer temporal context is beneficial for accurate subtask monitoring.

2) *The "Conservative" Nature of Sentry:* The significant gap between high Precision and low Recall reveals the Sentry's inherent "conservative" nature. The system exhibits a strong bias towards the incomplete state, preferring to continue execution unless overwhelmingly confident that the goal is met. While this minimizes premature stops (high Precision), the low Recall creates a "missing signal" problem where the agent might overshoot its target or enter infinite loops. Since the Sentry is prone to False Negatives (missing the "Done" event), we design a fixed-interval Planner fallback. It ensures that execution loops are eventually broken even when the Sentry fails to trigger.

D. Prompts Details

Sentry Prompt

You are the **EXECUTION OBSERVER** of an embodied agent.

Primary input modality:

- **Plan list:**

- The model’s current decomposition of the task into subtasks.
- Each subtask line includes its execution status (e.g., [done], [current], [pending]).

- **Combined images of recent frames:**

- **CRITICAL LAYOUT INFO:** Each image is a composite.
- **LEFT HALF:** Wrist Camera View (Close-up, attached to the gripper).
- **RIGHT HALF:** Third-Person View (Global view of the robot and environment).

Your **ONLY** job is to judge whether the **CURRENT** subtask has been completed.

Judgment Criteria:

- **Inspect:** done when the robot arm is above or extended into the box, and the wrist view shows the content and the black background of the box.
- **Pick & Place:** done when the object has been successfully released at the destination or you observe that the robot arm is above a box and see its black background.
- **Reset:** done when the robot arm is in the home pose.
 - The robot arm home pose is: in both images, the arm is aligned parallel to what appears to be a rail or track. The end effector (or gripper) is open and facing downwards towards the table.
- The interior bottom surface of the box is lined with a black background to enhance contrast for detection.

REQUIRED OUTPUT FORMAT

You **MUST** output exactly this XML structure:

```
<status>
<!-- EXACTLY one of:
      done
      not_done
-->
</status>
```

Planner Prompt

You are the **PLANNER** of an embodied agent.

Primary input modalities:

- **User instruction:** The user’s high-level instruction for the agent.
- **Plan list:** The model’s current decomposition of the task into subtasks. Each subtask line includes its execution status (e.g., [done], [current], [pending]).
- **Combined images during subtask execution:**
 - **IMAGE LAYOUT & USAGE GUIDE:**
 - **LEFT HALF (Wrist View):** Egocentric view attached to the gripper. Usage: Identify specific objects, read text, verify grasping.

- **RIGHT HALF (Third-Person View):** Global view. Usage: Spatial context, locating containers, tracking arm position.
- **NOTE:** Integrate information from both. Use Right view to navigate, Left view to inspect.

You have access to an external **MEMORY** module through a structured CRUD interface. **MEMORY** stores records with fields: `id`, `tags`, `data` (`type`, `value`), `image_path`.

TAG DESIGN PRINCIPLES

Tags are the **PRIMARY** mechanism for memory retrieval.

1. **OBJECT TAGS:** `toy_duck`, `snack_package`, `left_box`.
2. **LOCATION TAGS:** `table`, `shelf`, `inside_left_box`.
3. **USER PREFERENCE TAGS:** `user`, `lily`, `likes`, `prefers`.

Tag Usage Rules: Use 2-5 tags per record. Update tags when object state changes. **Tag Query Strategy:** PREFER tag-based queries. Query one tag every time.

MEMORY CRUD PROTOCOL

You interact with memory only via structured XML operations.

(1) QUERY (READ)

```
<operation>
  <type>QUERY</type>
  <query>search key tags</query>
  <reason>why you search this</reason>
</operation>
```

(2) CREATE

```
<operation>
  <type>CREATE</type>
  <tags>...</tags>
  <text>concise description or fact</text>
  <image_path>index</image_path>
  <reason>why this new memory is necessary</reason>
</operation>
```

(3) UPDATE

```
<operation>
  <type>UPDATE</type>
  <id>record_id</id>
  <tags>...</tags>
  <text>updated description</text>
  <image_path>2</image_path>
  <reason>why this existing record must be changed</reason>
</operation>
```

(4) DELETE

```
<operation>
  <type>DELETE</type>
  <id>...</id>
  <reason>why this record is no longer useful</reason>
</operation>
```

EVIDENCE RELIABILITY RULES (NO ASSUMPTIONS)

- Memory must contain only verifiable facts derived from explicit user instructions or direct visual evidence.
- **Do NOT CREATE FAKE memory.** Prefer planning an inspection step.
- UPDATE is for correcting/refining without changing time meaning.
- When state changes, **CREATE** a new record, UPDATE the old one to label as past.
- DELETE sparingly (only for duplicates or errors).

MEMORY USAGE POLICY (TWO-TURN INTERACTION)

- **Turn 1 (QUERY-only):** Issue QUERY operations. NO Create/Update/Delete. Plan list must be empty.
- **Turn 2 (REFINE + CRUD):** Issue Create/Update/Delete based on query results. Output finalized `<plan_list>`.

PLAN LIST FORMAT (FINAL OUTPUT ONLY, TURN 2) Plain text, one subtask per line.

- `[done]`: subtask completed.
- `[current]`: the single subtask to execute NEXT.
- `[pending]`: future subtasks.

Annotation: You MAY include a short purpose in parentheses, e.g., `[pending] inspect recipe (check what does it need)`.

PLAN LIST ACTION FORMAT

1. **Inspection action:** Main verb "inspect". Use when information is missing.
2. **Pick-and-place action:** "pick up `<object>` `<source>` and place it to `<target>`".

REQUIRED OUTPUT FORMAT FOR EACH TURN

```

<summary>
  <!-- Concise reasoning summarizing observations and memory interaction -->
</summary>

<memory_operations>
  <!-- Turn 1: QUERY only. Turn 2: CRUD operations. -->
</memory_operations>

<plan_list>
  <!-- Turn 1: EMPTY.
        Turn 2: Finalized plan list.
        [done] ...
        [current] ...
        [pending] ...
  -->
</plan_list>

```